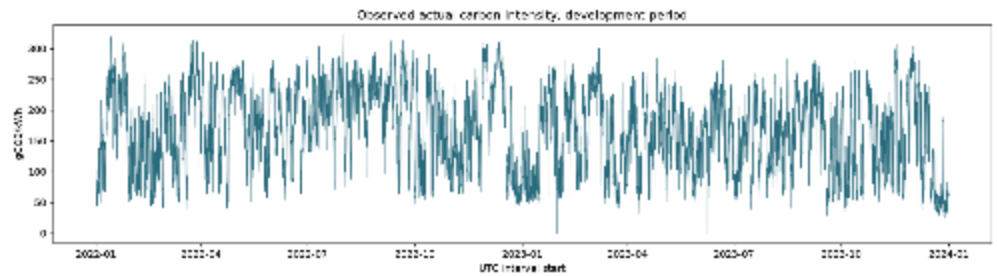
GridShift

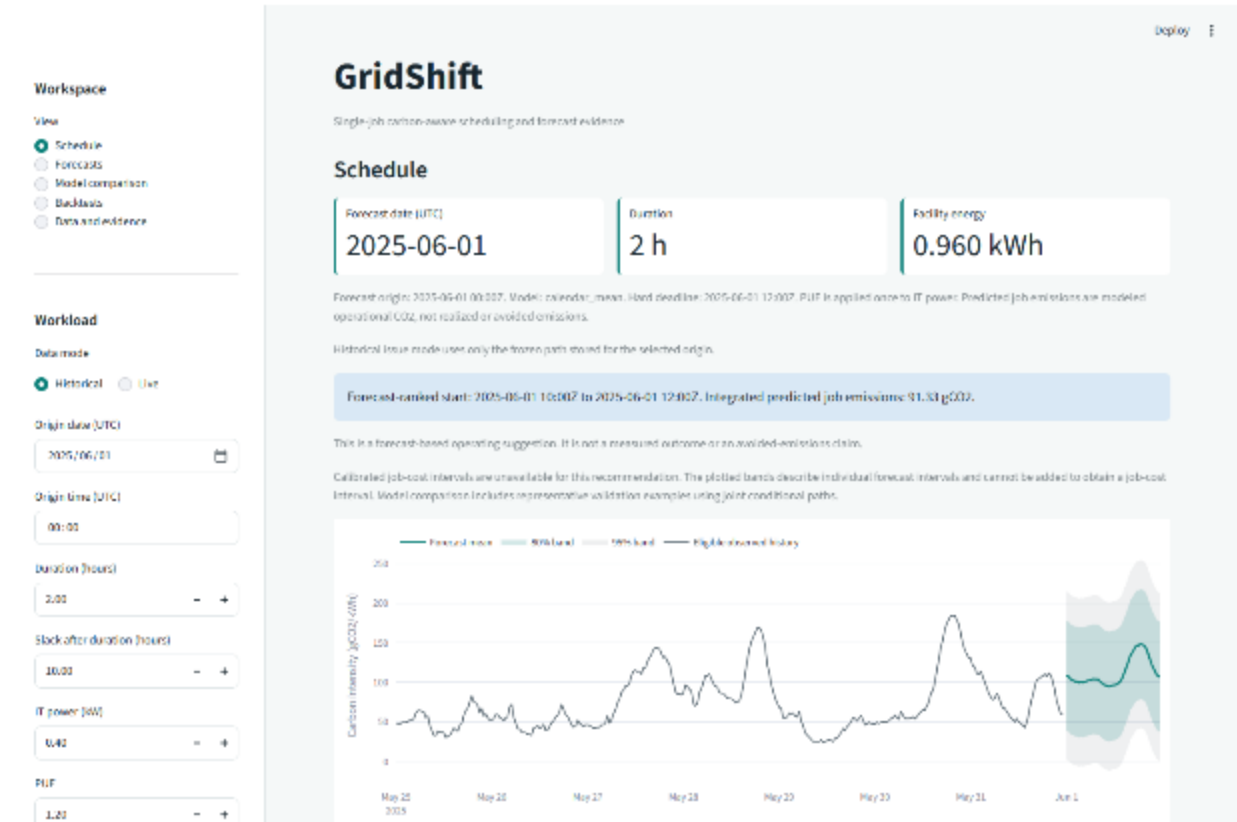
Carbon-aware GPU workload scheduling with classical time-series forecasts

- Study ID d2dee2d358735d7d
- Target: NESO estimated GB intensity, gCO2/kWh



Development observations

Dashboard and research scope

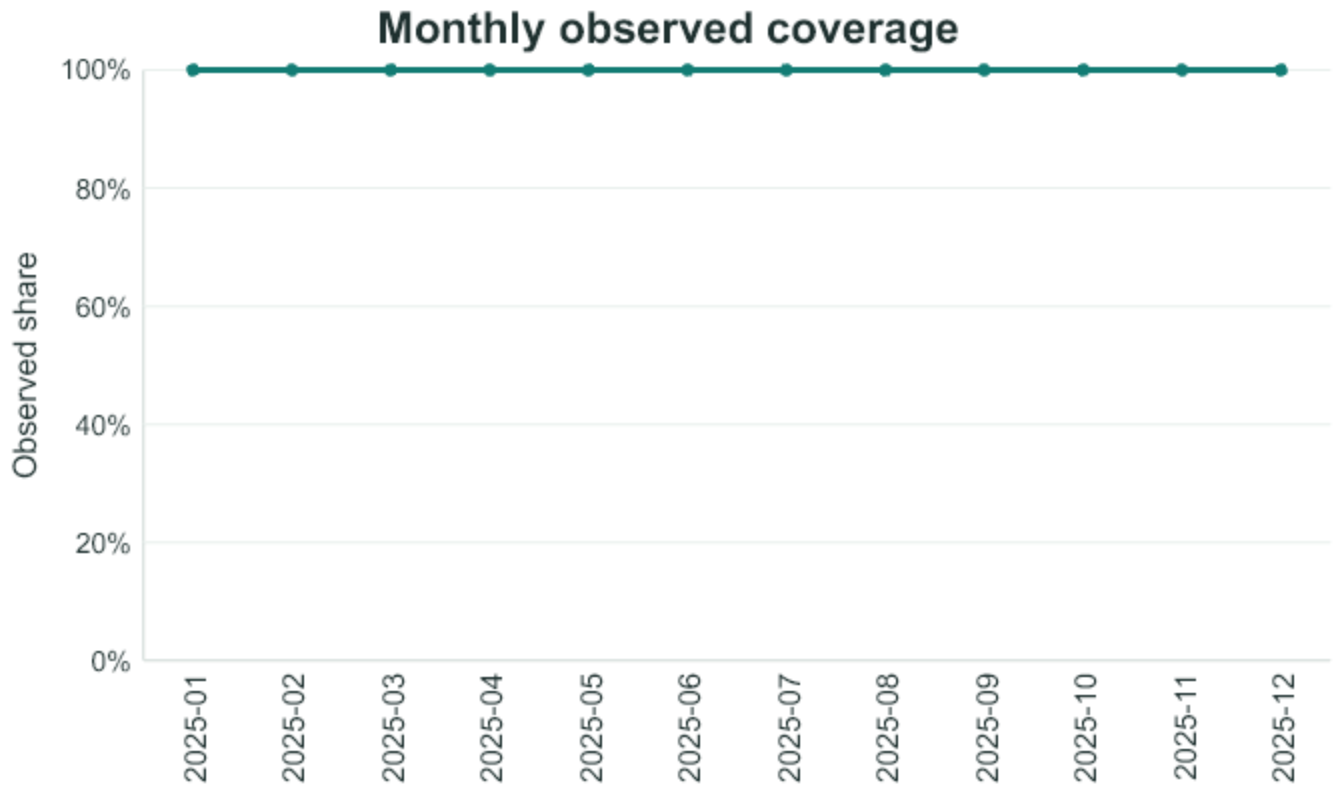


Verified local GridShift scheduling workspace

- Do eligible forecasts improve feasible GPU placement?
- Compare earliest execution and simple forecasts.
- Model operational CO2, not avoided grid emissions.
- Local recommendations are predictions, not measured savings.

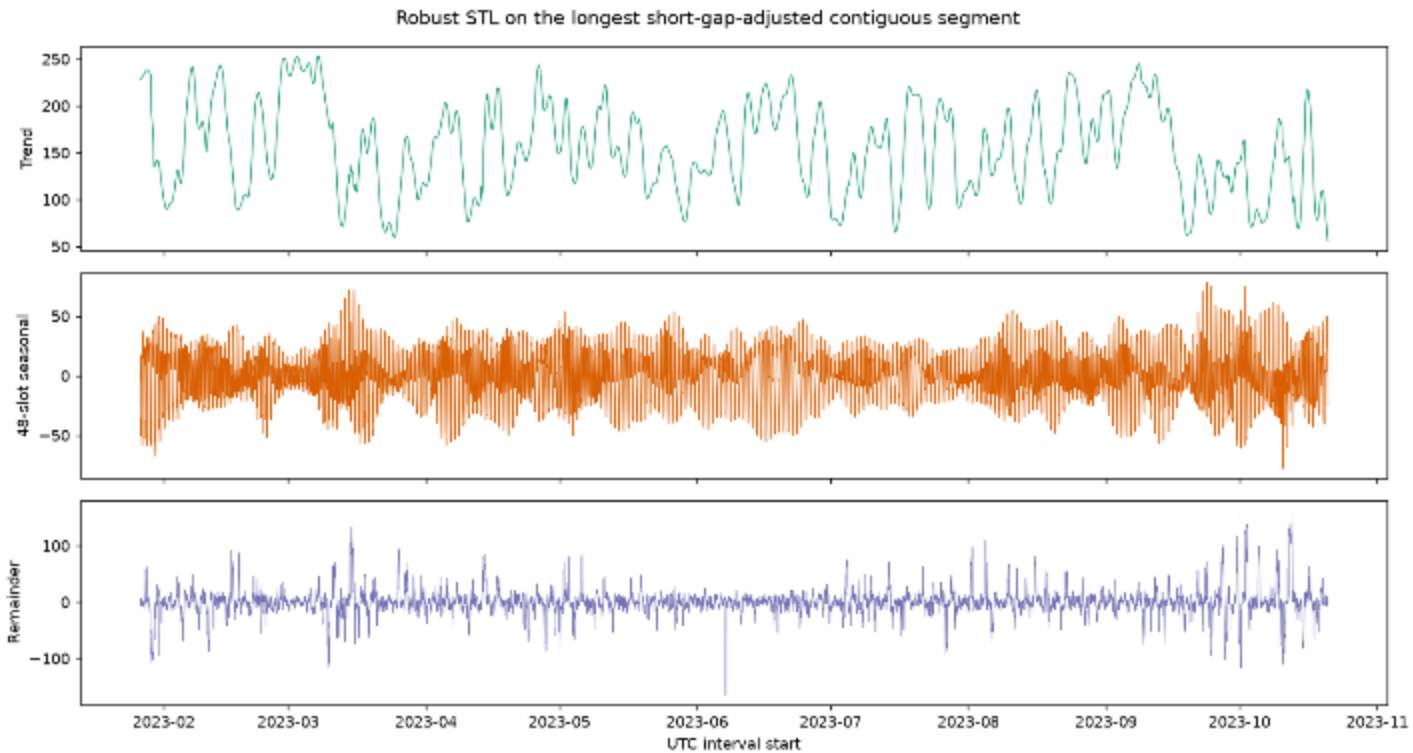
Data coverage and quality

- Observed slots: 69950 of 70128.
- Missing actuals remain missing; provider forecasts stay separate.
- The quality gate and monthly coverage are reported from stored artifacts.



Measure	Value
Slots	70128
Observed	69950
Missing	178
Overall coverage	0.9975
Lowest monthly coverage	0.9388
Lowest test-month coverage	1
Unique UTC intervals	true
Regular grid	true
Data gate passed	true
Assumed availability	true

Development diagnostics



Stored development-only robust STL components

- Diagnostic boundary: 2024-01-01T00:00:00+00:00.
- Development-only STL status: ok.
- ACF, periodogram, and stationarity checks guide candidates, not test conclusions.

Forecast model catalogue

- AR, MA, ARMA, ARIMA, and daily-seasonal SARIMA.
- Persistence, seasonal-naive, and calendar-mean controls.
- Failed fits and null outcomes remain visible in the source tables.

Role	Recorded choice
Forecast champion	SARIMA(1,0,1)x(1,0,0,48)
Strongest simple baseline	persistence
Scheduling model	calendar_mean
Selection boundary	Validation only

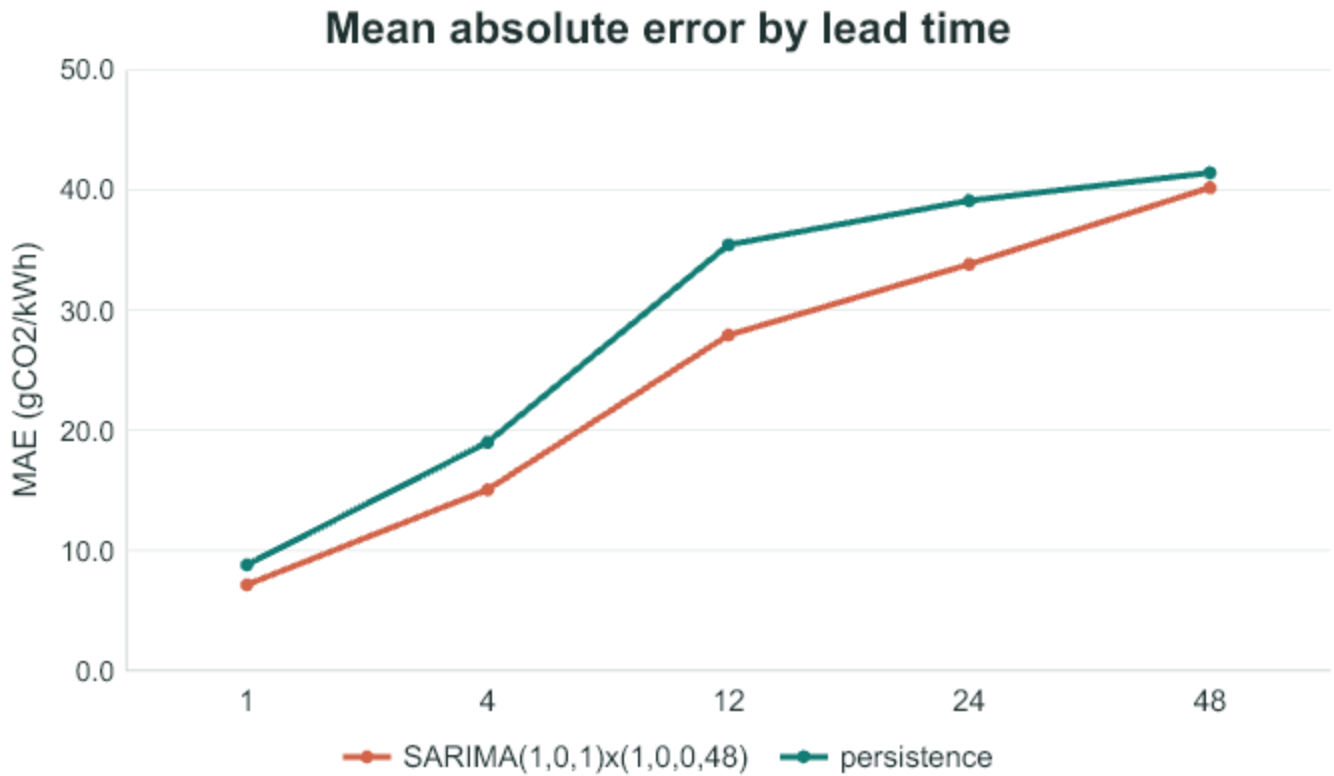
Evaluation design

- Chronological development, validation, and locked-test partitions.
- Common eligible targets and counts by lead time.
- Paired block uncertainty for the primary scheduling contrast.
- AIC and BIC compare only compatible likelihood groups.

Design item	Configured value
Validation	2024-01-01T00:00:00Z to 2025-01-01T00:00:00Z
Locked test	2025-01-01T00:00:00Z to 2026-01-01T00:00:00Z
Horizons	[1, 4, 12, 24, 48]
Bootstrap resamples	2000

Validation forecast outcomes

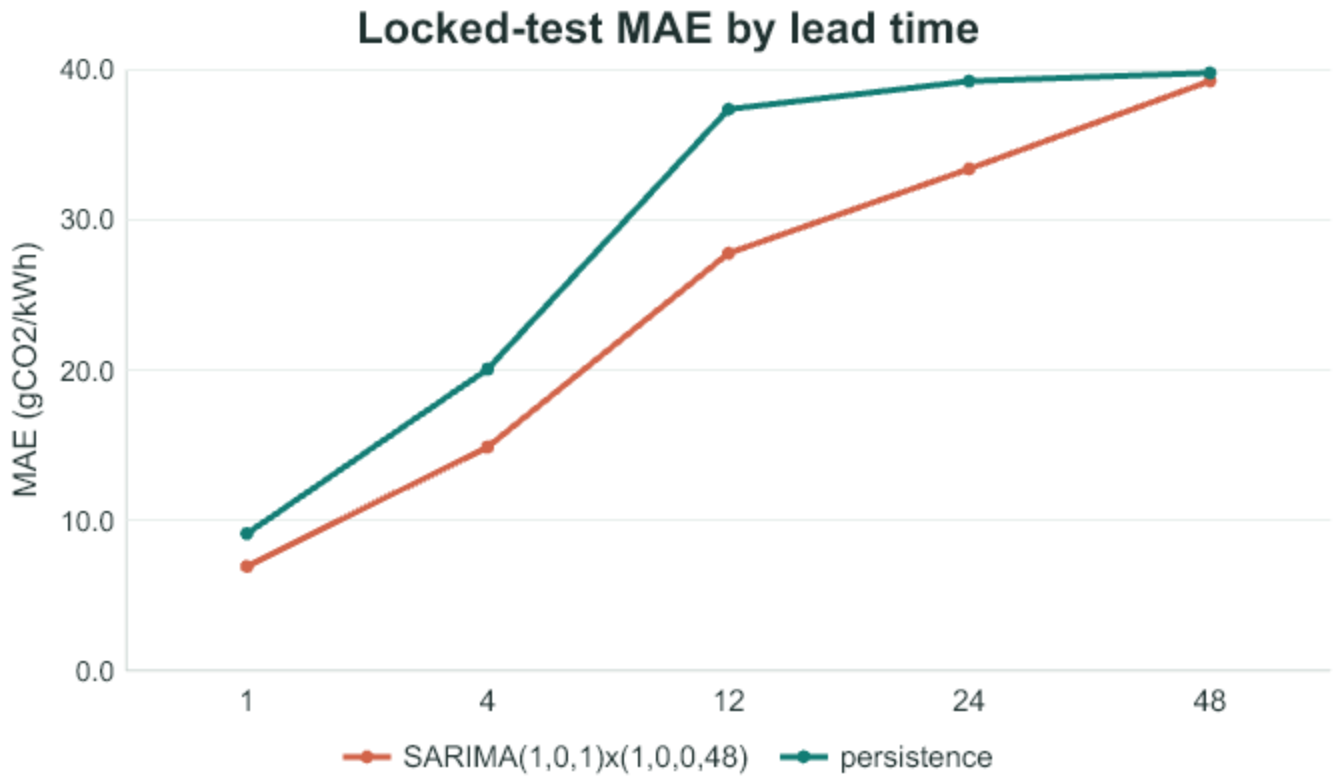
- Validation champion: SARIMA(1,0,1)x(1,0,0,48).
- Strongest simple baseline: persistence.
- The chart shows MAE by lead time only where the source table has a numeric value.



Model	Mean endpoint MAE
SARIMA(1,0,1)x(1,0,0,48)	24.823
persistence	28.754

Locked-test forecast outcomes

- Test rows report the validation-selected champion and simple baseline.
- The test does not change the selected model.
- Null metrics and zero-count groups remain unfilled.



Model	Mean endpoint MAE
SARIMA(1,0,1)x(1,0,0,48)	24.453
persistence	29.122

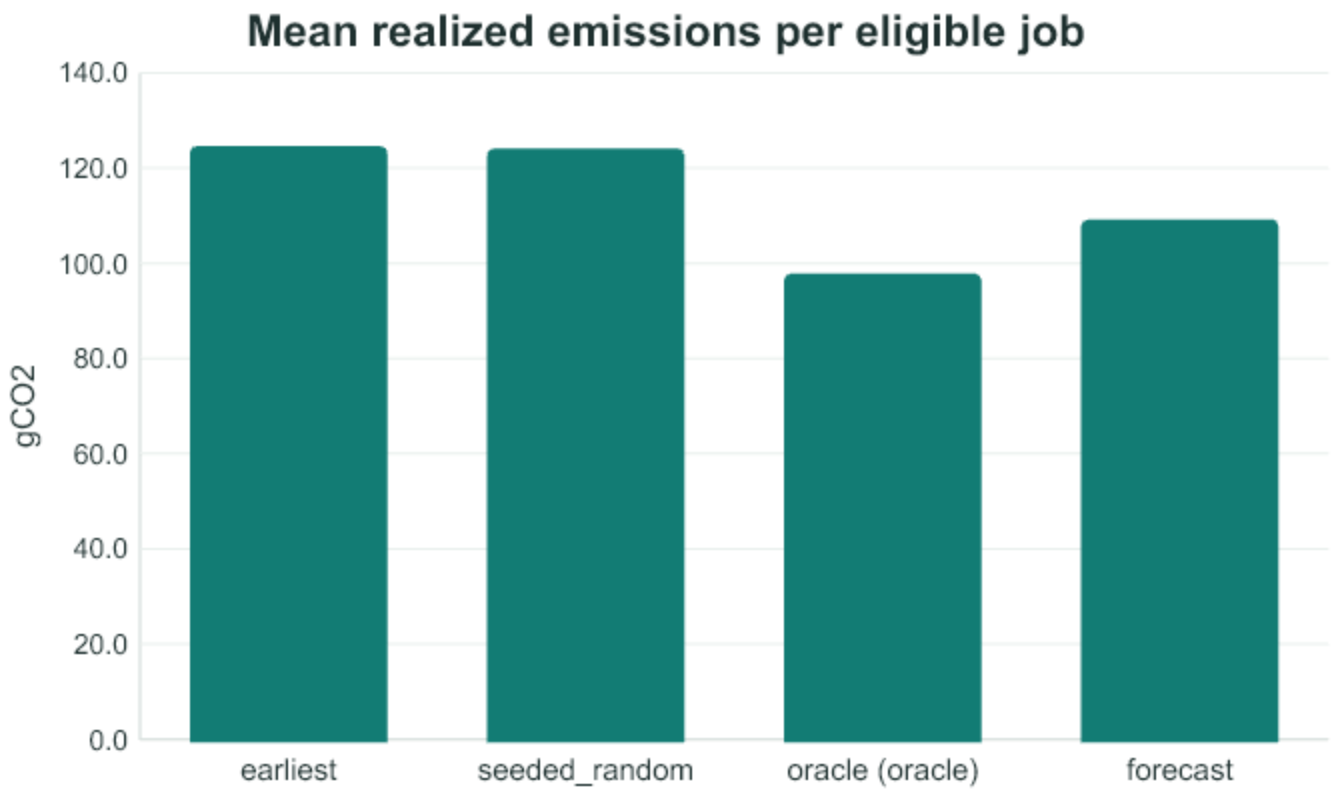
Scheduling formulation

- Minimize forecast intensity integrated over full job energy.
- Keep deadlines, capacity, and reservations as hard constraints.
- Use an oracle only as an unattainable opportunity bound.

Declared assumption	Value
Default duration (hours)	2
Default slack (hours)	10
Power (kW)	0.4
PUE	1.2
Primary model	calendar_mean

Primary scheduling outcomes

- calendar_mean: 12.414% modeled reduction against earliest execution.
- 95% paired block interval: 10.951% to 13.828%.



Policy	Mean gCO2	Delta %
earliest	124	NA
oracle	97.22	21.61
forecast	108.6	12.41

Queue service and emissions

- Table values are arithmetic means across the three frozen seeds.
- Matched emissions differences are descriptive; accepted work differs.
- Lower emissions from rejected work are not savings.

Load	Slack h	Reject early %	Reject selected %	Matched delta %
0.3	6	0.48518	2.373	6.7787
0.3	12	0.023624	0.36428	8.531
0.6	6	3.6671	9.7095	4.3856
0.6	12	0.41371	3.9073	5.1506
0.85	6	9.7584	17.737	3.571
0.85	12	3.8893	10.701	3.6974

Backup equations and failures

- Job CO_2 = sum(overlap hours x IT kW x PUE x intensity).
- Choose the minimum integrated forecast cost; obey release, deadline, and capacity.
- SARIMA uses AR, MA, and period-48 seasonal lag polynomials on differenced levels.
- Missing or stale forecasts disable carbon ranking; failed fits retain explicit fallbacks.
- Rejected queue work is not savings. Marginal intervals are not joint job-cost guarantees.
- Frozen calendar_mean timing benefit and classical forecast skill are separate findings.